STUDYLAB HUB · COLLEGE CALCULUS I & AP CALCULUS · UNIT 1 (CHAPTER 1)

Unit 1: Limits, Continuity & Derivatives

One-Sided Limits · Algebraic Indeterminate Forms (0/0) · Squeeze Theorem · IVT · Limit Definition of the Derivative

College Calculus I & AP Calculus — Unit 1 (Chapter 1): Limits, Continuity & Definition of the Derivative

Independent, original study guide for College Calculus I & AP Calculus AB/BC (informed by OpenStax Calculus Volume 1, CC BY 4.0).

1. Unit 1 Roadmap (Chapter 1)

Section	Core Mathematical Definition	Key Technique		
1.1 · Understanding Limits	$\lim_{x \to c} f(x) = L \iff \lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = L$	Check left-hand and right-hand limits independently on piecewise functions and absolute value expressions $\frac{	x-a	}{x-a}$
1.2 · Algebraic Limit Laws ($0/0$ Indeterminate Forms)	Factor & cancel, conjugate rationalization, common denominators	Never leave $= \frac{0}{0}$ as a final statement on FRQs (causes an immediate linkage error deduction!)		
1.3 · Trigonometric Limits & Squeeze Theorem	$\lim_{x \to 0} \frac{\sin(kx)}{kx} = 1$ and $\lim_{x \to 0} \frac{1 - \cos x}{x} = 0$	Bound oscillating terms like $x^2 \sin(1/x)$ between $-x^2 \le x^2\sin(1/x) \le x^2$		
1.4 · Continuity at a Point ($x = c$)	Three-part test: (1) $f(c)$ exists, (2) $\lim_{x \to c} f(x)$ exists, (3) $\lim_{x \to c} f(x) = f(c)$	Classify discontinuities as removable (hole), jump ($\text{LHL} \ne \text{RHL}$), or infinite (vertical asymptote)		
1.5 · Infinite Limits & Asymptotes	Vertical asymptote at $x=a$ if $\lim_{x \to a^\pm} f(x) = \pm\infty$; Horizontal asymptote if $\lim_{x \to \pm\infty} f(x) = L$	For rational functions with $\sqrt{x^2 + \dots}$ as $x \to -\infty$, remember $\sqrt{x^2} =	x	= -x$
1.6 · Intermediate Value Theorem (IVT)	If f is **continuous** on $[a,b]$ and k is between $f(a)$ and $f(b)$, there exists $c \in (a,b)$ with $f(c) = k$	Always state explicitly: *"Because $f(x)$ is continuous on $[a,b]$..."* before invoking IVT		
1.7 · Limit Definition of the Derivative	$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ and $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$	Recognize limit expressions that are disguised derivatives (e.g., $\lim_{h \to 0} \frac{\ln(e + h) - 1}{h} = \left.\frac{d}{dx}[\ln x]\right\|_{x=e} = \frac{1}{e}$)		

2. The 4 Essential Algebraic Limit Moves ($\frac{0}{0}$)

1. **Factoring & Canceling**:

$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - 2x - 3} = \lim_{x \to 3} \frac{(x-3)(x+3)}{(x-3)(x+1)} = \frac{3+3}{3+1} = \frac{6}{4} = \frac{3}{2}$$

1. **Rationalizing Radicals with the Conjugate**:

$$\lim_{x \to 0} \frac{\sqrt{x+4} - 2}{x} \cdot \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} = \lim_{x \to 0} \frac{(x+4) - 4}{x(\sqrt{x+4} + 2)} = \lim_{x \to 0} \frac{1}{\sqrt{x+4} + 2} = \frac{1}{4}$$

1. **Clearing Complex Fractions**:

$$\lim_{x \to 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2} = \lim_{x \to 2} \frac{2 - x}{2x(x - 2)} = \lim_{x \to 2} \frac{-(x - 2)}{2x(x - 2)} = -\frac{1}{4}$$

1. **Special Trigonometric Limits**:

$$\lim_{x \to 0} \frac{\sin(5x)}{\sin(3x)} = \lim_{x \to 0} \left(\frac{\sin(5x)}{5x} \cdot \frac{3x}{\sin(3x)} \cdot \frac{5}{3}\right) = 1 \cdot 1 \cdot \frac{5}{3} = \frac{5}{3}$$

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3. Worked Exam Problems

Problem 1 (Piecewise Continuity Parameter Solve)

Find the constants a and b that make $f(x)$ continuous everywhere:

$$f(x) = \begin{cases} 2x + 1, & x \le 1 \\ ax + b, & 1 < x < 3 \\ x^2 + 2, & x \ge 3 \end{cases}$$

- **Solution**:
- Continuity at $x = 1$: $\lim_{x \to 1^-} (2x+1) = \lim_{x \to 1^+} (ax+b) \implies 3 = a + b$.
- Continuity at $x = 3$: $\lim_{x \to 3^-} (ax+b) = \lim_{x \to 3^+} (x^2+2) \implies 3a + b = 11$.
- Subtracting the first equation from the second: $2a = 8 \implies \mathbf{a = 4}$, and $\mathbf{b = -1}$.

Problem 2 (Horizontal Asymptotes with Radicals)

Evaluate $\lim_{x \to -\infty} \frac{3x - 2}{\sqrt{4x^2 + 5}}$.

- **Solution**: For $x < 0$, $\sqrt{x^2} = |x| = -x$. Divide numerator and denominator by $x = -\sqrt{x^2}$:

$$\lim_{x \to -\infty} \frac{3x - 2}{-x\sqrt{4 + 5/x^2}} = \lim_{x \to -\infty} \frac{3 - 2/x}{-\sqrt{4 + 5/x^2}} = \frac{3}{-2} = -\frac{3}{2}$$